

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

CR23532- Cyber and Digital Forensics

LAB MANUAL

2023 REGULATION

FIFTH SEMESTER

III YEAR

COLLEGE VISION

- To be an institution of excellence in Engineering, Technology and Management, Education & Research.
- To provide competent and ethical professionals with a concern for society.

COLLEGE MISSION

- To impart quality technical education imbibed with proficiency and humane values.
- To provide right ambience and opportunities for the students to develop into creative, talented and globally competent professionals.
- To promote research and development in technology and management for the benefit of the society.

DEPARTMENT VISION

To promote highly ethical and innovative cybersecurity professionals through excellence in teaching, training, and research.

DEPARTMENT MISSION

- To produce globally competent cybersecurity experts, motivated to learn emerging technologies and be innovative in solving real-world security challenges.
- To foster research activities among students and faculty that enhance societal security and resilience.
- To impart moral and ethical values in their profession, ensuring a strong commitment to cybersecurity ethics and responsible practices.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of networking, security, data mining, web technology, mobile communication and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry requirements.

PROGRAMME OUTCOMES (POs)

PO1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, naturalsciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leaderin diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leaderin a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage inindependent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Computer Science and Engineering Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze, and develop computer programs in the areas related to algorithms, system software, web design, machine learning, data analytics, networking, and cybersecurity for efficient design of computer-based systems of varying complexity. Familiarity and practical competence with a broad range of programming languages, open-source platforms, and cybersecurity tools and practices.

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational tasks, model real-world problems using appropriate data structures and suitable algorithms, and address cybersecurity challenges. Understanding of standard practices and strategies in software project development, incorporating secure coding practices and using open-ended programming environments to deliver quality and secure products.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and provide solutions to new ideas, with a particular emphasis on cybersecurity challenges. Inculcating a passion for higher studies, creating innovative career paths to be an entrepreneur, and

evolving as an ethically and socially responsible computer science professional with strong cybersecurity expertise.

Subject Code	Subject Name (Lab Oriented Theory course)	Category	L	T	P	C
CR23532	Cyber and Digital Forensics	PC	3	0	2	4

Objectives:	
•	Have an introduction into the process of Cyber Forensics.
•	Understand the Environment of forensics.
•	Learn process of collecting evidences.
•	Gain working knowledge of analyzing evidences using tools.
•	Learn other sources of evidences and its future challenges.

UNIT I	DIGITAL FORENSICS PROCESS	9
Forensic Science, Digital Forensics, Digital Evidence, Digital Forensics Process – Identification, Collection, Examination, Analysis, Presentation Phases. Cyber Crime Law- International Legal Framework of Cybercrime Law, Digital Crime, Investigation Methods for Collecting Digital Evidence.		
UNIT II	**FORENSICS ENVIRONMENT**	**9**
Hardware and Software Environments – Storage Devices, Operating System, File Systems, Metadata, Locating evidence in file systems-Password security, Encryption, and Hidden files. Case study –linking the evidence to the user, Data Analysis using forensics tool ILookIX		
UNIT-III	**COLLECTING EVIDENCES**	**9**
Use of Digital Evidence, File Metadata and Correlation with Other Evidence, Technical Complexities of Digital Evidence. Data carving, Date and time problems, Physical Acquisition and Safekeeping of Digital Evidence. Forensic Imaging Processes. Case Study – IXImager, Understanding ASB container		
UNIT IV	**ANALYZING EVIDENCES**	**9**
Selecting and Analyzing Digital Evidence - Locating digital evidence, categorizing files, eliminating superfluous files, The Event Analysis tool, Cloud Analysis tool, The Lead Analysis tool, Volume Shadow Copy analysis tools, Validating the Evidence. Case study – illustrating the recovery of deleted evidence held in volume shadows.		
UNIT-V	**OTHER SOURCES OF EVIDENCES**	**9**
Windows and Other Operating Systems as Sources of Evidence, Examining Browsers, E-mails, Messaging Systems, and Mobile Phones, Internet and Cloud. -Challenges in Digital Forensics.		
	Total Contact Hours :	**45**

List of Experiments	
1	Study of Computer Forensics and different tools used for forensic investigation
2	Live Forensics Case Investigation using Autopsy
3	Recover Deleted Files using Forensics Tools
4	Find Last Connected USB on your system (USB Forensics)
5	View Last Activity of Your PC
6	Extraction of Browser Artifacts
7	Comparison of two Files for forensics investigation by Compare IT software
8	Collecting Email Evidence in Victim PC
9	Study the steps for hiding and extract any text file behind an image file/ Audio file using Command Prompt.

10	Extract Exchangeable image file format (EXIF) Data from Image Files using Exifreader Software	
	Contact Hours :	**30**
	Total Contact Hours :	**75**

Course Outcomes:	
On completion of course you will be able to	
•	Identify the need for cybercrime investigation.
•	Understand the hardware and software components responsible for seeking evidence.
•	Have knowledge on the techniques used for collecting evidences.
•	Analyse the evidence through suitable tools.
•	Examine other sources of evidences.

Suggested Activities:	
•	Assignment problems.
•	Class presentation/Discussion

Textbooks:	
1.	Richard Boddington, Practical Digital Forensics, First Edition, , PACKT publishing, 2016

Reference Books (s)/Web links:	
1.	John R.Vacca, Computer Forensics, Second Edition, Cengage Learning, 2005.
2.	Richard E.Smith, Internet Cryptography, Third Edition, Pearson Education, 2008.
3.	Marjie T.Britz, Computer Forensics and Cyber Crime: An Introduction, Third Edition, Prentice Hall, 2013.

CO-PO-PSO Matrices of course

PO/PSO CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CR23532.1	3	2	-	-	1	2	1	3	2	1	-	-	2	-	1
CR23532.2	3	2	2	1	3	1	-	-	-	-	-	-	3	2	-
CR23532.3	2	2	2	1	3	1	-	-	-	1	-	-	3	3	-
CR23532.4	1	3	3	3	3	-	-	1	2	2	-	-	2	3	2
CR23532.5	2	3	2	3	3	2	1	1	1	-	-	1	1	3	3
Average	2.2	2.4	2.2	2	2.6	1.5	1	1.6	1.6	1.3	-	1	2.2	2.7	2